

CAPE ST JAMES, BC, Canada

WMO: 711070

Lat: 51.9358N

Lon: 131.0158W

Elev: 89

StdP: 100.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.9	-0.1	-9.6	1.7	-0.7	-7.3	2.0	0.8	25.2	5.2	22.8	4.6	8.3	60	0.915

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	19.2	15.3	17.8	14.7	16.6	14.1	16.4	18.4	15.5	17.2	14.7	16.2	4.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.1	17.3	14.7	10.5	16.2	14.0	10.1	15.3	46.1	18.5	43.5	17.3	41.4	16.3	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.9	16.8	DB	-3.6	22.5	3.6	2.0	-6.2	24.0	-8.3	25.2	-10.3	26.3	-12.9	27.8	(4)
(5)			WB	-5.9	18.2	3.2	1.3	-8.2	19.2	-10.1	20.0	-11.9	20.8	-14.2	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	5.6	5.5	5.6	7.4	9.4	11.6	13.5	14.4	13.3	10.3	7.4	5.7
	DBStd	3.77	2.63	2.12	2.05	1.74	1.79	1.77	1.39	1.39	1.53	1.66	2.08	2.35
	HDD10.0	742	136	126	136	81	31	3	0	0	0	17	80	132
	HDD18.3	3347	394	359	393	328	276	203	151	122	151	251	329	391
	CDD10.0	438	0	0	1	3	14	51	108	137	100	24	1	0
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH23.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.3	9.8	9.0	8.9	8.3	7.3	7.3	7.1	6.9	7.0	8.8	9.6	9.9
Precipitation	PrecAvg	1599	180	142	134	118	93	83	65	81	122	203	195	183
	PrecMax	2092	304	279	228	237	201	165	180	169	279	315	322	304
	PrecMin	1145	21	25	39	40	34	18	7	6	21	106	98	88
	PrecStd	227	57	52	48	46	44	39	38	45	56	51	58	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	10.7	11.6	15.0	17.3	21.3	20.6	21.1	19.7	16.1	11.9	10.4
		MCWB	9.7	8.0	8.3	8.5	12.0	14.6	15.7	17.4	15.0	12.5	10.6	9.3
	2%	DB	10.0	9.3	9.8	12.9	15.1	17.4	18.1	19.2	17.9	14.1	11.0	9.6
		MCWB	9.2	7.7	7.7	8.7	10.8	13.5	15.5	16.1	14.2	12.1	10.0	8.7
	5%	DB	9.2	8.6	8.9	11.1	13.5	15.4	16.7	17.8	16.4	13.2	10.5	9.0
		MCWB	8.3	7.3	7.3	8.1	10.4	12.8	14.6	15.3	13.6	11.7	9.6	8.2
	10%	DB	8.4	8.0	8.1	9.7	12.0	14.0	15.7	16.6	15.3	12.5	10.0	8.5
		MCWB	7.6	6.8	6.9	7.5	9.7	12.1	13.9	14.6	13.3	11.1	9.2	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.2	9.6	10.7	13.3	15.7	17.2	18.1	16.2	13.8	11.4	9.9
		MCDB	10.3	9.9	10.5	13.0	16.2	19.5	19.6	20.4	18.3	14.4	11.5	10.1
	2%	WB	9.0	8.2	8.7	9.6	11.9	14.2	15.8	16.8	15.1	12.9	10.5	9.1
		MCDB	9.3	8.9	9.5	11.3	14.4	16.6	17.6	18.9	16.7	13.5	10.7	9.3
	5%	WB	8.2	7.6	7.9	8.8	10.9	13.3	15.0	15.9	14.3	12.2	10.0	8.5
		MCDB	8.5	8.3	8.6	10.3	13.0	15.1	16.7	17.4	15.6	12.9	10.2	8.8
	10%	WB	7.7	7.1	7.2	7.9	10.1	12.5	14.2	15.1	13.8	11.5	9.5	7.9
		MCDB	8.0	7.8	8.0	9.5	11.8	14.0	15.5	16.4	14.9	12.2	9.9	8.3
Mean Daily Temperature Range	5% DB	MDBR	2.4	3.1	3.5	3.9	4.6	4.4	4.5	4.4	4.0	3.3	2.9	2.7
		MCDBR	2.3	3.6	4.3	6.3	7.1	7.0	6.4	6.6	5.9	4.4	2.9	2.6
		MCWBR	2.3	3.1	3.5	4.6	5.2	5.0	4.6	4.7	4.3	3.4	2.8	2.8
	5% WB	MCDBR	2.4	3.2	3.5	5.5	6.5	6.5	6.3	6.3	5.0	3.7	2.5	2.4
		MCWBR	2.4	2.9	3.2	4.3	4.9	4.9	4.7	4.6	3.9	3.3	2.7	2.9
Clear-Sky Solar Irradiance	taub	0.319	0.325	0.351	0.388	0.385	0.369	0.368	0.370	0.369	0.349	0.327	0.310	
	taud	2.401	2.480	2.415	2.290	2.363	2.459	2.516	2.514	2.471	2.438	2.361	2.357	
	Ebn at Noon	686	803	844	847	868	885	880	860	816	754	659	623	
	Edh at Noon	60	73	93	118	115	106	98	94	88	75	61	53	
All-Sky Solar Radiation	RadAvg	0.66	1.52	2.60	4.10	5.12	5.41	5.19	4.44	3.22	1.80	0.88	0.55	
	RadStd	0.08	0.23	0.28	0.20	0.37	0.45	0.48	0.37	0.29	0.19	0.08	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page